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1.0 Introduction

DSP manipulates different types of signals with the intention of **filtering, measuring, or compressing and producing analog signals**. Analog signals differ by taking information and translating it into electric pulses of **varying amplitude**, whereas digital signal information is translated into binary format where each bit of data is represented by two distinguishable amplitudes. Another noticeable difference is that analog signals can be represented as sine waves and digital signals are represented as square waves. DSP can be found in almost any field, whether it's oil processing, sound reproduction, radar and sonar, medical image processing, or telecommunications-- essentially any application in which signals are being compressed and reproduced.

Digital Signal Processors (DSP) take real-world signals like voice, audio, video, temperature, pressure, or positions that have been digitized and then mathematically manipulate them. A DSP is designed for performing mathematical functions like "add", "subtract", "multiply" and "divide" very quickly.



Analog Signal



Digital Signal

Signals need to be processed so that the information that they contain can be displayed, analyzed, or converted to another type of signal that may be of use. **In the real-world, analog products detect signals such as sound, light, temperature or pressure and manipulate them. Converters such as an Analog-to-Digital converter then take the real-world signal and turn it into the digital format of 1's and 0's. From here, the DSP takes over by capturing the digitized information and processing it. It then feeds the digitized information back for use in the real world. It does this in one of two ways, either digitally or in an analog format by going through a Digital-to-Analog converter. All of this occurs at very high speeds.**

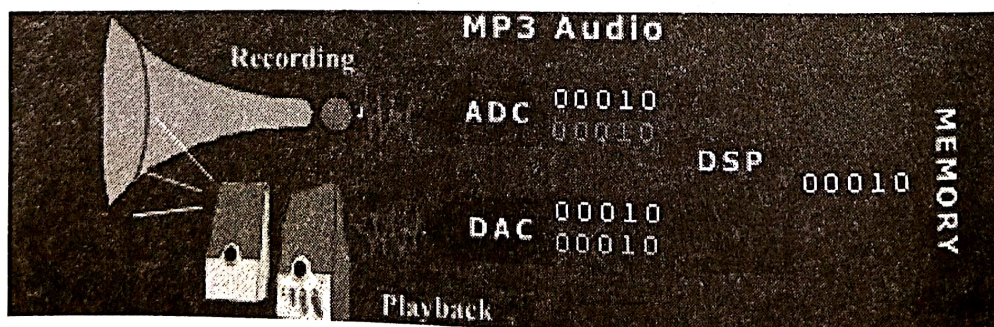


Figure 1.1: Process of recording and playing back of the signal using ADC and DAC

To illustrate this concept, the figure 1.1 above shows how a DSP is used in an MP3 audio player. During the recording phase, analog audio is input through a receiver or other source.

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This analog signal is then converted to a digital signal by an analog-to-digital converter and passed to the DSP. The DSP performs the MP3 encoding and saves the file to memory.

During the playback phase, the file is taken from memory, decoded by the DSP and then converted back to an analog signal through the digital-to-analog converter so it can be output through the speaker system. In a more complex example, the DSP would perform other functions such as volume control, equalization and user interface.

A DSP's information can be used by a computer to control such things as security, telephone, home theater systems, and video compression. Signals may be compressed so that they can be transmitted quickly and more efficiently from one place to another (e.g. teleconferencing can transmit speech and video via telephone lines). Signals may also be enhanced or manipulated to improve their quality or provide information that is not sensed by humans (e.g. echo cancellation for cell phones or computer-enhanced medical images). Although real-world signals can be processed in their analog form, processing signals digitally provides the advantages of high speed and accuracy.

Because it's programmable, a DSP can be used in a wide variety of applications. You can create your own software or use software provided by ADI and its third parties to design a DSP solution for an application.

1.1 A DSP contains these key components:

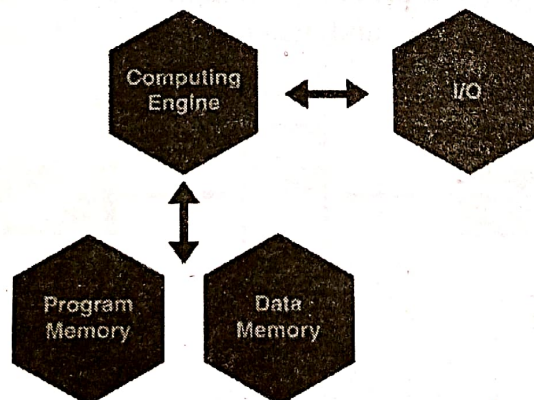


Figure 1.2: Key components of DSP

- **Program Memory:** Stores the programs the DSP will use to process data
- **Data Memory:** Stores the information to be processed
- **Computing Engine:** Performs the math processing, accessing the program from the Program Memory and the data from the Data Memory
- **Input/Output:** Serves a range of functions to connect to the outside world

1.2 Basic Elements of Digital Signal Processing

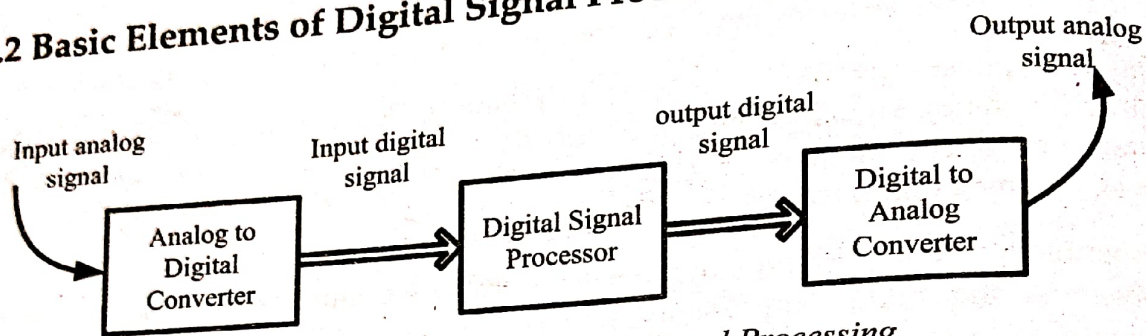


Figure 1.3: Elements of Digital Signal Processing

ADC & DAC

Electric equipment is heavily used in nearly every field. Analog to Digital Converters (ADC) and Digital to Analog Converters (DAC) are essential components for any variation of DSP in any field. These two converting interfaces are necessary to convert real world signals to allow for digital electronic equipment to pick up any analog signal and process it. Take a microphone for example: the ADC converts the analog signal collected by an input to audio equipment into a digital signal that can be outputted by speakers or monitors. While it is passing through the audio equipment to the computer, software can add echoes or adjust the tempo and pitch of the voice to get a perfect sound. On the other hand, DAC will convert the already processed digital signal back into the analog signal that is used by audio output equipment such as monitors. Below is a figure showing how the previous example works and how its audio input signals can be enhanced through reproduction, and then outputted as digital signals through monitors.

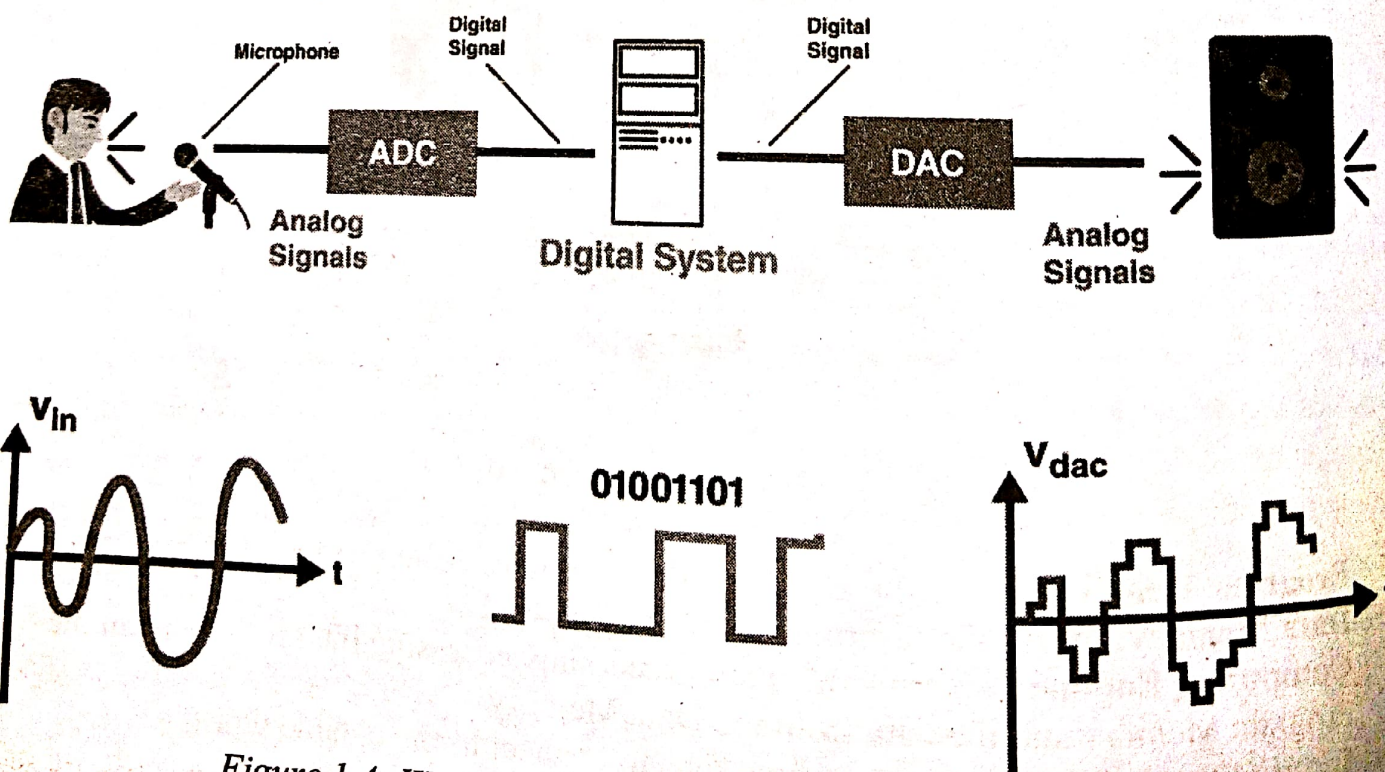


Figure 1.4: Working of Digital system using ADC and DAC

1.3 Applications of DSP

There are numerous variants of a digital signal processor that can execute different things, depending on the application being performed. Some of these variants are audio signal processing, audio and video compression, speech processing and recognition, digital image processing, and radar applications. The difference between each of these applications is how the digital signal processor can filter each input. There are five different aspects that vary from each DSP: clock frequency, RAM size, data bus width, ROM size, and I/O voltage. All of these components really are just going to affect the arithmetic format, speed, memory organization, and data width of a processor.

Digital signal processing is heavily used in day-to-day operations, and is essential in recreating analog signals to digital signals for many purposes.

The main applications of DSP are:

1. Audio signal processing,
2. Audio compression,
3. Digital image processing,
4. Video compression,
5. Speech processing,
6. Speech recognition,
7. Digital communications,
8. Radar and Sonar signal processing,
9. Seismology
10. Biomedicine.

Specific examples are:

Speech compression and transmission in digital mobile phones, room correction of sound in hi-fi and sound reinforcement applications, weather forecasting, economic forecasting, seismic data processing, analysis and control of industrial processes, medical imaging such as CAT scans and MRI, MP3 compression, computer graphics, image manipulation, hi-fi loudspeaker crossovers and equalization, and audio effects for use with electric guitar amplifiers.

1.4 Advantages of Digital Signal Processing

Digital signal processing has following advantages:

1. Digital signal processing operations can be changed by changing the program in digital programmable system, *i.e.*, flexible systems.
2. Better control of accuracy in digital systems compared to analog systems.
3. Digital signals are easily stored on magnetic media such as magnetic tape without loss of quality of reproduction of signal.
4. Digital signals can be processed off line, *i.e.*, these are easily transported.
5. Sophisticated signal processing algorithms can be implemented by DSP method.
6. Digital circuits are less sensitive to tolerances of component values.
7. Digital systems are independent of temperature, ageing and other external parameters.
8. Digital circuits can be reproduced easily in large quantities at comparatively lower cost.
9. Cost of processing per signal in DSP is reduced by time-sharing of given processor among a number of signals.
10. Processor characteristics during processing, as in adaptive filters can be easily adjusted in digital implementation.
11. Digital system can be cascaded without any loading problems.

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Example 1.1 Consider the analog signal,

$x_a(t) = 2 \cos(3000\pi t) + 3 \sin(4000\pi t) + 7 \cos(6000\pi t)$. If the sampling rate is 8000 samples per second and quantized at 8-bits, find:

- Discrete values at any two points
- Quantization errors at those points

Solution:

Sampling rate (F_s) = 8000 samples/sec

Discrete time signal obtained after sampling,

$$x(n) = 2 \cos(3000\pi n/F_s) + 3 \sin(4000\pi n/F_s) + 7 \cos(6000\pi n/F_s)$$

$$\text{Or, } x(n) = 2 \cos(3000\pi n/8000) + 3 \sin(4000\pi n/8000) + 7 \cos(6000\pi n/8000)$$

$$x(n) = 2 \cos(1.178n) + 3 \sin(1.571n) + 7 \cos(2.356n)$$

i). Now for discrete values at any two points, consider $n=1$ and $n=2$

At $n=1$

$$x(1) = 9.076$$

At $n=2$

$$x(2) = 9.13$$

The signal is quantized at 8-bits therefore, $n=8$ and this implies that the system is Pulse Code Modulation (PCM)

$$\text{Quantization error for PCM } (Q_e) = \frac{\Delta^2}{12}$$

Where, Δ represents step size

$$\Delta = \frac{x_{\max}}{2^{n-1}}$$

For $n=1$, $x_{\max} = 9.076$ similarly, for $n=2$, $x_{\max} = 9.13$

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For $n=1$,

$$\Delta = \frac{x_{max}}{2^{n-1}} = \frac{9.076}{2^7} = 0.071$$

$$Q_e = \frac{\Delta^2}{12} = 4.19 \times 10^{-4}$$

For $n=2$,

$$\Delta = \frac{x_{max}}{2^{n-1}} = \frac{9.13}{2^7} = 0.0713$$

$$Q_e = \frac{\Delta^2}{12} = 4.24 \times 10^{-4}$$

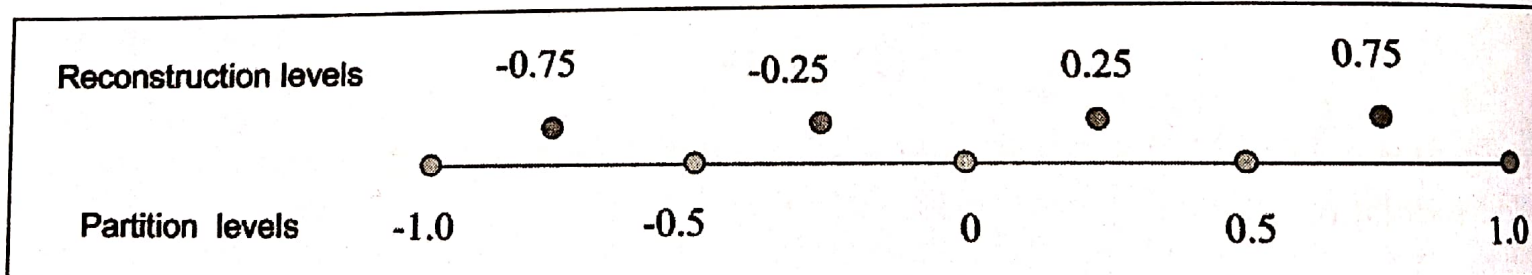
Example 1.2 Determine and illustrate the partition levels and reconstruction levels of a uniform quantizer in the range of $(-1, 1)$ with 4 levels.

Solution:

The entire data range is $-1+1=2$

$N=4$, $Q=2/4=0.5$

The partition and reconstruction levels are illustrated below.



1. For the sequence $\{0.2, -0.3, -0.7, 0.8\}$,
 - a. Determine the quantized sequence using the uniform quantizer of Prob. 1. Also determine the mean square quantization error.
 - b. Determine the binary stream corresponding to the quantized sequence.

Solution:

- a. The quantized sequence is $\{0.25, -0.25, -0.75, 0.75\}$.

The quantization error is

$$\sigma_q^2 = \frac{1}{N} \sum_n (x(n) - \hat{x}(n))^2$$

$$= \frac{1}{4} ((0.2 - 0.25)^2 + (0.3 - 0.25)^2 + (0.7 - 0.75)^2 + (0.8 - 0.75)^2) = 0.0025$$

- b. $\{10, 01, 00, 11\}$

There are 4 quantized levels, so we need 2 bits to represent each. The above solution is based on the following mapping of quantized levels to codewords:

$-0.75 \rightarrow "00", -0.25 \rightarrow "01", 0.25 \rightarrow "10", 0.75 \rightarrow "11"$.

Example 1.3 Consider the analog signal:

$$x_a(t) = 3 \cos(2000\pi t) + 5 \sin(6000\pi t) + 10 \cos(12000\pi t)$$

- a) What is the Nyquist rate for this signal?
- b) Suppose that the signal is sampled at the rate of $F_s = 5000$ samples per second, what is the discrete-time signal obtained after sampling?
- c) What is the analog signal we can reconstruct from the samples?

Solution:

- a) The frequencies existing in the above signal are

$$F_1 = 1000 \text{ Hz}$$

$$F_2 = 3000 \text{ Hz}$$

$$F_3 = 6000 \text{ Hz}$$

Thus, $F_{max} = 6000 \text{ Hz}$

According to the sampling Theorem

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$$F_s > 2F_{max} = 12000\text{Hz}$$

Hence, Nyquist rate = 12000Hz

- b) The discrete-time signal of the signal sampled at 5000 samples per second is

$$\begin{aligned}x(n) &= 3\cos\left(\frac{2000\pi}{5000}\right)n + 5\sin\left(\frac{6000\pi}{5000}\right)n + 10\cos\left(\frac{12000\pi}{5000}\right)n \\&= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(\frac{3}{5}\right)n + 10\cos 2\pi\left(\frac{6}{5}\right)n \\&= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(1 - \frac{2}{5}\right)n + 10\cos 2\pi\left(1 + \frac{1}{5}\right)n \\&= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(-\frac{2}{5}\right)n + 10\cos 2\pi\left(\frac{1}{5}\right)n\end{aligned}$$

Finally, we obtain

$$x(n) = 13\cos 2\pi\left(\frac{1}{5}\right)n - 5\sin 2\pi\left(\frac{2}{5}\right)n$$

Since $F_s = 5000\text{Hz}$, the folding frequency is $\frac{F_s}{2} = 2500\text{Hz}$. This is the maximum frequency that can be represented uniquely by the sampled signal

- c) Since only the frequency components at 1000 Hz and 2000 Hz are present in the sampled signal, the analog signal that can be recovered is

$$y(t) = 13\cos 2\pi(1000)t - 5\sin 2\pi(2000)t$$

which is obviously different from the original signal $x(t)$. This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.